AAKARSH ANAND

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Machine learning researcher working on foundation models and statistical genetics for health data

EDUCATION

Ph.D., Computer Science, UCLA · Major field: AI · Advisor: Sriram Sankararaman — 2024 – 2029 (expected)
M.S., Computer Science, UCLA (earned during the Ph.D.) — 2026
B.S., Computer Science, UCLA — 2020 – 2024

RESEARCH EXPERIENCE

Graduate Student Researcher, Sankararaman Lab, UCLA (undergraduate researcher 2022–24) — Jan 2022 – present

Foundation models for health data

- **Wearable foundation models.** Inertia-1 (NeurIPS 2026 main track, co-first author, with Yuzhe Yang's lab), an open study of pretraining on wrist accelerometry, with 5M–100M-parameter models in six architectures, 18.2M hours from 115K+ people, and 15 evaluation datasets. Worked on much of the pretraining and evaluation, and showed that combining sensors consistently beats any single sensor.
- **Disease prediction and genetics.** A follow-up journal paper (in prep., co-first author) applies the models to diagnosis, prognosis, and disease progression in five cohorts and studies the genetics of the learned traits, with early work pairing them with LLMs.
- **Genetics-aware training.** Developing a loss that pushes a model's learned traits to be heritable and genetically correlated with disease.

Statistical genetics at biobank scale

- **Gene–gene interactions.** Co-developed scalable tests for interactions between variants (QuadKAST, co-first author, and FAME) and ran their calibration and power simulations. Now extending it to rare disease, supported by an Anthropic AI for Science grant.
- **Gene–environment interactions.** Helped develop ENGINE, which learns which combinations of exposures change genetic effects, and REGENA (second author, in revision), which separates trait scale and amplification from real differences in genetic effects.
- **Genetic effects across genes, cells, and tissues.** Ran real-data analyses for FLEX, which estimates the heritability of rare variants gene by gene. Also building a single-cell model of genetic effects across cell types and a genetic-correlation atlas across 53 tissues.

PUBLICATIONS

* equal contribution

- Inertia-1: An Open Exploration of Wearable Motion Foundation Models
  Z. Xu*, **A. Anand***, S. Jiang*, C. Zhuang, Z. Shuai, S. Sankararaman, Y. Yang · **NeurIPS 2026 main track** · also an oral at the GenAI for Health workshop · code
- A biobank-scale method for learning modulators of gene-environment interaction underlying human complex traits from multiple environmental exposures
  Z. Liu, A. Ramteke, **A. Anand**, A. Gorla, M. Jeong, S. Sankararaman · **Genome Research 2026** · selected from RECOMB 2026
- A biobank-scale test of marginal epistasis reveals genome-wide signals of polygenic interaction effects
  B. Fu, A. Pazokitoroudi, Z. Shi, A. Kar, A. Xue, **A. Anand**, P. Anand, et al. · **Nature Genetics 2025**
- A scalable adaptive quadratic kernel method for interpretable epistasis analysis in complex traits
  B. Fu*, P. Anand*, **A. Anand***, J. Mefford, S. Sankararaman · **Genome Research 2024** · selected from RECOMB 2024
- Disentangling phenotype scale, amplification, and effect heterogeneity underlying polygenic GxE architecture of complex traits
  Z. Liu, **A. Anand**, X. Qie, M. Costantino, M. Jeong, A. Gorla, N. Zaitlen, A. Dahl, S. Sankararaman · in revision at **Nature Genetics** · code
- Comprehensive gene heritability estimation reveals the genetic architecture of rare coding variants underlying complex traits
  Z. Liu, B. Fu, M. Jeong, P. Anand, **A. Anand**, et al. · bioRxiv 2025 · in revision at **Nature Genetics**

AWARDS AND FUNDING

Anthropic AI for Science Grant, Rare Disease Research — 2026
NIH T32 Predoctoral Fellowship, UCLA Genomic Analysis Training Program — 2026 – 2027
Warren Alpert Computational Biology and AI Network Fellowship, UCLA — 2024

TEACHING

Teaching Assistant, UCLA — 2025 – 2026

- AI for Computational Genomics (Spring 2025, ~120 students). Co-wrote three new assignments for its AI redesign.
- Machine Learning Applications in Genetics (Spring 2026) and Machine Learning Applications in Biomedicine (Winter 2025).

EARLIER EXPERIENCE

Software Engineering Intern, LabCorp · Snakemake pipelines for high-volume sequencing data — Jun – Sep 2023
Research Assistant, Boutros Lab, UCLA · cancer genomics pipelines, co-author of Metapipeline-DNA — 2021 – 2022

SKILLS

Machine learning: PyTorch, Transformers, CNNs, RNNs, self-supervised pretraining, multimodal fusion, model evaluation, LLMs
Statistics and genetics: method-of-moments estimation, variance components, heritability, genetic correlation, GWAS, simulations
Engineering and data: Python, NumPy/pandas, SLURM, Git, Claude Code · UK Biobank, NHANES, OneK1K, GTEx